

Níckolas de Aguiar Alves (he/him)

 PhD candidate in physics

 Federal University of ABC

 Portuguese (native), English (fluent), Italian (basic)

 Brazilian

 aguialvalves.com

 nick@aguialvalves.com

 alves.nickolas@ufabc.edu.br

Education

2023–
Sep.



Doctor in Physics

Federal University of ABC (UFABC), Santo André, São Paulo, Brazil

Advisor: André G. S. Landulfo 

2021–2023
Feb. Apr.



Master in Physics

Federal University of ABC (UFABC), Santo André, São Paulo, Brazil

Advisor: André G. S. Landulfo 

Thesis: *Nonperturbative Aspects of Quantum Field Theory in Curved Spacetime*

2017–2020
Feb. Dec.



Bachelor in Physics

Institute of Physics, University of São Paulo (IFUSP), São Paulo, São Paulo, Brazil

Advisor: João C. A. Barata 

Appointments

2026–2027
Aug. Aug.



Visiting PhD Candidate

Scuola Normale Superiore di Pisa (SNS), Pisa, Toscana, Italy

Supervisor: Guilherme L. Pimentel 

Funded by the São Paulo Research Foundation (FAPESP) Grant No. 2026/02509-8

Publications

Research Articles

- 1 N. Aguiar Alves and A. G. S. Landulfo. *Sound as a gauge theory and its infrared triangle*. 2026. arXiv: 2512.15796 [hep-th]. Submitted.
- 2 N. Aguiar Alves and A. G. S. Landulfo. “Null infinity as a Killing horizon”. *Phys. Rev. D* **112.6**, 065017 (2025). arXiv: 2504.12514 [gr-qc].
- 3 N. Aguiar Alves, A. G. S. Landulfo, and B. A. Costa. “Positive mass in general relativity without energy conditions”. *Phys. Rev. D* **111.4**, 044027 (2025). arXiv: 2408.00154 [gr-qc].

Review Articles

- 1 N. Aguiar Alves. “Lectures on the Bondi–Metzner–Sachs group and related topics in infrared physics”. *Eur. Phys. J. C* **86.5**, 507 (2026). arXiv: 2504.12521 [gr-qc].

Pedagogical Articles

- 1 N. Aguiar Alves. “The unexpected effectiveness of the Picard maneuver: Seeing three images of the same superluminal spaceship”. *Am. J. Phys.* arXiv: 2608.23624 [physics.pop-ph]. Forthcoming.
- 2 C. C. Rodrigues Evangelista and N. Aguiar Alves. “Using thermodynamics to learn gravitational wave physics”. *Eur. J. Phys.* **47.2**, 025604 (2026). arXiv: 2602.21261 [gr-qc].

Essays and Short Contributions

- 1 N. Aguiar Alves. “Physics as a branch of poetry”. *Am. J. Phys.* **94.7** (2026), p. 507.
- 2 N. Aguiar Alves and B. A. Costa. *The Measure of a Mass*. 2025. arXiv: 2503.18963 [gr-qc]. Pre-published.

Theses

- 1 N. Aguiar Alves. “Nonperturbative Aspects of Quantum Field Theory in Curved Spacetime”. MSc thesis. Santo André, Brazil: Federal University of ABC, 2023. xxiv, 152 pp. arXiv: [2305.17453 \[gr-qc\]](#).

Other Works

- 1 N. Aguiar Alves. “Quantum Field Theory in Curved Spacetime. An Introduction”. Unpublished lecture notes. 2023.

In Preparation

- 1 N. Aguiar Alves, A. G. S. Landulfo, and A. D. Pereira. *Nonperturbative renormalization group flow for a particle detector*. In preparation.
- 2 N. Aguiar Alves, C. C. Rodrigues Evangelista, G. J. Olmo, and D. Rubiera-Garcia. *Platypus stars: Exotic compact objects supported by vacuum pressure*. In preparation.

Funding

2026–	📖	PhD Scholarship – Research Internship Abroad Explorations with Twistors and Cosmological Correlators São Paulo Research Foundation (FAPESP) Grant No. 2026/02509-8 Advisor in Brazil: André G. S. Landulfo  Supervisor abroad: Guilherme L. Pimentel 
2025–	📖	PhD Scholarship The Sky as a Killing Horizon and Other Topics on the Infrared Structure of Gravity São Paulo Research Foundation (FAPESP) Grant No. 2025/05161-0 Advisor: André G. S. Landulfo 
2023–2025	📖	PhD Scholarship Coordenação de Aperfeiçoamento do Pessoal de Ensino Superior (CAPES) Granted to pursue my PhD after placing first in UFABC's admission ranking
2021–2023	📖	MSc Scholarship The Functional Renormalization Group in Quantum Field Theory in Curved Spacetimes São Paulo Research Foundation (FAPESP) Grant No. 2021/07372-7 Advisor: André G. S. Landulfo 
2021	📖	MSc Scholarship Coordenação de Aperfeiçoamento do Pessoal de Ensino Superior (CAPES) Granted to pursue my MSc after placing first in UFABC's admission ranking
2019–2020	📖	BSc Scholarship Hyperbolic Equations São Paulo Research Foundation (FAPESP) Grant No. 2019/12158-4 Advisor: João C. A. Barata 

Academic Service

Conferences Organized

- | | | |
|------|---|---|
| 2026 | 📖 | Thermality in Quantum Field Theory in Curved Spacetimes
Main organizer
Promoted by the ICTP South American Institute for Fundamental Research |
| 2023 | 📖 | Golden Wedding of Black Holes and Thermodynamics
Main organizer |

Academic Service (continued)

Schools Organized

- 2024–  [São Paulo School in Gravitational Physics](#)
Main organizer
Part of [ICTP Physics Without Frontiers](#) (second edition onward)
- 2018–2021  [Jayme Tiomno School in Theoretical Physics](#)
Founder and organizer of the first three editions

Peer Review

- 2026–  Reviewer for [Physical Review D](#)
 Reviewer for the [American Journal of Physics](#)

Scientific Societies

- 2025–  [Brazilian Physical Society](#)
- 2024–  [International Society for Quantum Gravity](#)

Teaching Experience

Minicourses and Workshops

- 2025  Introduction to Black Hole Physics
[II São Paulo School in Gravitational Physics](#)
Minicourse for early undergraduate students
- 2024  Infrared Symmetries of General Relativity
[I São Paulo School in Gravitational Physics](#)
Minicourse for advanced undergraduate and early graduate students
- 2023  Quantum Field Theory in Curved Spacetime
[Golden Wedding of Black Holes and Thermodynamics](#)
Minicourse for early graduate students
- 2021  Algebraic Methods of Theoretical Physics
[III Jayme Tiomno School in Theoretical Physics](#)
Minicourse on group theory in physics for first-year undergraduate students
- 2020  An Introduction to $\text{\LaTeX}$
[XV Oceanography Thematic Week](#)
Oceanographic Institute, University of São Paulo
- 2018  An Introduction to $\text{\LaTeX}$
[I Jayme Tiomno School in Theoretical Physics](#)

Teaching Assistant

- 2025  Topics in Celestial Holography
Institute of Physics, University of São Paulo
Graduate-level course
-  Advanced Topics in General Relativity
Institute of Physics, University of São Paulo
Graduate-level course
- 2024  Electromagnetic Phenomena
Federal University of ABC
Undergraduate-level course covering introductory electrodynamics
- 2021  Classical Mechanics II
Federal University of ABC
Undergraduate-level course covering analytical mechanics

Teaching Experience (continued)

- 2019  **Mathematical Physics I**
Institute of Physics, University of São Paulo
Undergraduate-level course covering introductory Fourier analysis

Participation in Panels

- 2026  **VI ICTP-SAIFR Summer School for Young Physicists**
Panel with graduate students in physics

Outreach and Miscellanea

-  **VI ICTP-SAIFR Summer School for Young Physicists**
Tutored a course on black hole physics for high school students
- 2025  **International Young Physicists Tournament (Brazil)**
Juror for the final stage of the 2025 edition of the IYPT Brazil
-  **V ICTP-SAIFR Summer School for Young Physicists**
Tutored a course on black hole physics for high school students
- 2018–  **Physics Stack Exchange**
Active contributor to the Physics Stack Exchange Q&A website
Most prolific contributor to the [qft-in-curved-spacetime](#) tag

Awards

- 2025  “Outstanding Recognition” award at the **UFABC Physics Week**
- awarded to 12 out of 45 participants based on talks about their research progress

Press Coverage

- 1 M. von Hippel. *Physicist does the math on Star Trek’s “Picard maneuver”. The trick from Next Generation would be even more effective in real life*. Ars Technica, 2026. On **N. Aguiar Alves**. “The unexpected effectiveness of the Picard maneuver: Seeing three images of the same superluminal spaceship”. *Am. J. Phys.* arXiv: [2608.23624](#) [[physics.pop-ph](#)]. Forthcoming.

Research Presentations

Contributed Talks

- 1 *Sound as a gauge theory and its infrared triangle*. **Simons–Balseiro School and Workshop on Advanced Aspects of QFT**, Instituto Balseiro, May 18–27, 2026.
- 2 *Quantum Gravity and the Night Sky*. **UFABC Physics Week**, Federal University of ABC, Oct. 14–17, 2025.
- 3 *Positive mass in general relativity without energy conditions*. **XLVII Paulo Leal Ferreira Congress**, Institute of Theoretical Physics, São Paulo State University, Oct. 29–Nov. 1, 2024.
- 4 *Nonperturbative renormalization group flow for a particle detector*. **Golden Wedding of Black Holes and Thermodynamics**, Online, Dec. 4–8, 2023.

Poster Presentations

- 1 *You can teach gravitational wave physics using thermodynamics*. **Third generation gravitational wave detectors: The view from Latin America**, Principia Institute, June 29–July 3, 2026.
- 2 *We can use the renormalization group to study the nonperturbative behavior of particle detectors*. **Simons–Balseiro School and Workshop on Advanced Aspects of QFT**, Instituto Balseiro, May 18–27, 2026.

- 3 *Linear sound has asymptotic symmetries.* **XI QCD Meets Gravity Conference**, Principia Institute, Dec. 15–19, 2025.
- 4 *Linear sound has asymptotic symmetries.* **XI QCD Meets Gravity School**, Institute of Theoretical Physics, São Paulo State University, Dec. 8–12, 2025.
- 5 *Stars with extreme pressures can mimic black holes.* **XLVIII Paulo Leal Ferreira Congress**, Institute of Theoretical Physics, São Paulo State University, Oct. 14–17, 2025.
- 6 *Killing horizons naturally define asymptotic symmetries.* **First International Latin American Conference on Gravitational Waves**, National Institute for Space Research, Sept. 15–19, 2025.
- 7 *Killing horizons naturally define asymptotic symmetries.* **VI Siembra-HoLAGrav Young Frontiers Meeting at ICTP-SAIFR**, Institute of Theoretical Physics, São Paulo State University, June 30–July 11, 2025.
- 8 *Killing horizons naturally define asymptotic symmetries.* **Program on Anomalies, Topology and Quantum Information in Field Theory and Condensed Matter Physics**, Principia Institute, June 16–27, 2025.
- 9 *Negative masses are unstable and we don't need energy conditions to prove it.* **Witnessing Quantum Aspects of Gravity in a Lab**, Principia Institute, Sept. 23–27, 2024.
- 10 *Nonperturbative aspects of quantum field theory in curved spacetime.* **XLVI Paulo Leal Ferreira Congress**, Institute of Theoretical Physics, São Paulo State University, Oct. 24–27, 2023.
- 11 *Nonperturbative aspects of quantum field theory in curved spacetime.* **V Jayme Tiomno Physics School**, Institute of Physics, University of São Paulo, Sept. 4–8, 2023.
- 12 *Nonperturbative aspects of quantum field theory in curved spacetime.* **Interfaces Between Quantum and Classical Statistical Mechanics**, Institute of Mathematics and Statistics, University of São Paulo, July 24–28, 2023.

Flash Talks

- 1 *Nonperturbative renormalization group flow for a particle detector.* **School on Phase Transitions and Gravitational Waves**, International Institute of Physics, Federal University of Rio Grande do Norte, Mar. 4–8, 2024.
- 2 *Nonperturbative renormalization group flow for a particle detector.* **Quantum Spacetime and the Renormalization Group**, Sant'Elmo Beach Hotel, Oct. 2–6, 2023.

Seminars

- 1 *Could frogs teach us something about quantum gravity?* University of Naples Federico II, Oct. 21, 2026.
- 2 *Could frogs teach us something about quantum gravity?* Scuola Normale Superiore, Oct. 5, 2026.
- 3 *Sound as a gauge theory and its infrared triangle.* **Simons Collaboration on Celestial Holography Zoom Seminars**, Online, Mar. 10, 2026.
- 4 *Null infinity as a Killing horizon.* Center for Relativity and Cosmology, Troy University, Aug. 27, 2025.
- 5 *The sky as a Killing horizon.* **Siembra-HoLAGrav Future of Physics**, Online, May 8, 2025.

I have also given dozens of group seminars at the Federal University of ABC and at the University of São Paulo, which I omit here for conciseness.